PROJECT ROADMAP

Refitting exoplanet transits

NASA Exoplanet Archive for targets and published benchmarks, Lightkurve for TESS photometry, and allesfitter for transit fits and posterior uncertainties.

Research question

For a confirmed transiting planet, how precisely can TESS photometry recover the radius ratio, orbital period and transit epoch, and how sensitive are those estimates to data preparation and model choices?

WEEK 1	**Set up and shortlist** Run a bundled allesfitter example in a pinned Python environment. Export NASA archive rows and shortlist 3-5 targets.
WEEK 2	**Choose a target and benchmark** Inspect MAST coverage. Freeze one primary target, one backup and one published reference. Save sector and transit counts.
WEEK 3	**Prepare the photometry** Clean and normalize each sector. Verify time conventions, protect transit windows and export fit-ready CSVs.
WEEK 4	**Fit one sector** Plot the initial model. Fit one sector, inspect the transits and residuals, and revise the baseline or data cuts if needed.
WEEK 5	**Fit all selected sectors** Share the transit parameters across sectors. Save posterior samples and check that repeated runs agree.
WEEK 6	**Test uncertainty sensitivity** Rerun with alternative detrending and noise choices. Run a small injection-recovery check and quantify parameter shifts.
WEEK 7	**Compare with published values** Align definitions and epochs, document data and prior overlap, and assemble the comparison table and plots.
WEEK 8	**Reproduce and write up** Rerun from saved inputs. Deliver the report, data manifest, configurations, posterior summaries and figures.

Weeks 9-12: Add a second target or deepen validation of the first. The extension schedule is on page 5.

Select and prepare the data

Start with a confirmed, short-period planet with several complete transits and good out-of-transit coverage.

1. Save a reproducible archive snapshot

Query the Planetary Systems table (**PS**) via TAP. The starter query selects default rows. Save the query, download date and raw CSV. Retrieve all PS rows for your finalist and select a single, documented literature solution for comparison. [1-2]

```
SELECT pl_name, hostname, tic_id, pl_refname, pl_orbper,
       pl_tranmid, pl_trandur, pl_ratror, pl_trandep,
       pl_tsystemref, sy_tmag, ttv_flag
FROM ps
WHERE default_flag = 1 AND tran_flag = 1
  AND pl_controv_flag = 0 AND pl_orbper BETWEEN 0.5 AND 5
```

For the reference row, retain the matching **err1**, **err2** and limit flags where available. Preserve the archive signs in the snapshot. Period and epoch are in days, duration in hours and depth in percent. Record missing or limit-only entries explicitly. [1]

2. Verify the data before committing to a target

Search MAST with **lk.search_lightcurve(tic_id, mission="TESS", author="SPOC", exptime=120)**. Check the TIC ID, author, exposure, sectors and valid transit coverage before downloading. Aim for at least two sectors and five complete transits in total. Favor clear ingress and egress, low crowding and a simple timing model. WASP-121 b is a candidate to screen. [3-4]

3. Clean and normalize by sector

Use SPOC PDCSAP flux with a recorded quality mask. Drop nonfinite flux or times and nonfinite or nonpositive flux errors. Check gaps and duplicates, and use one product per time range. Scale flux and its errors together. If further detrending is needed, mask transits and fit the out-of-transit trend by segment, scaling both flux and errors by that trend. Keep the unflattened series for comparison. [3,5-6]

Week 3 checkpoint

Save product IDs, sectors, cadence, quality rules and kept points. Plot each transit with its mask overlaid. Confirm the ephemeris lines up before fitting. Use a backup target if the available coverage or blending is unsuitable.

Build the allesfitter fit

Test one sector first. Then fit all selected sectors together with a shared transit model.

1. Put the timestamps and epoch on the same system

For TESS, convert to full BJD_{TDB} with **lc.time.tdb.jd**. BTJD is BJD_{TDB} minus 2,457,000. Verify the published epoch standard before comparison. Phase the data and check transit centers visually. Place the fitted epoch near the center of the observations and record the reference transit number. [1,5,7]

2. Build a fit directory per run

Start with the official photometry example. Include **settings.csv**, **params.csv** and one CSV per sector named to match its instrument entry (for example, **TESS_S10.csv**). Export sorted **time, flux, flux_err** columns with relative flux near one. Use unbinned data for inference. Set exposure integration for longer cadences or whenever exposure is appreciable relative to ingress. [7-8]

3. Set the transit model and priors

Fit **b_rr** (planet-to-star radius ratio), **b_rsuma** (sum of radii divided by semimajor axis), **b_cosi**, **b_period** and **b_epoch**. Use physically valid bounds and justify limb-darkening priors for the TESS band. Use a circular baseline only with literature support. Give each sector its own baseline and white-noise scale. Use local transit windows with adequate baseline on both sides if fitting transit-only. [8-9]

4. Avoid reusing the benchmark as a tight prior

Use archive values as starting guesses and to locate transits. Adopt priors broad enough to contain the plausible posterior. If a short time baseline requires an external period prior, document it and report the resulting precision as conditional on that prior. PDCSAP includes a crowding correction. Apply any residual dilution term only when supported by independent evidence. [6,9]

5. Run and validate sampling

Run **allesfitter.show_initial_guess(datadir)** before sampling. For nested sampling use **ns_fit** and **ns_output**. Repeat with a fresh seed or more live points and check agreement of key quantiles. Review corner plots, prior boundaries, residuals by sector and posterior predictive transit plots. Keep configurations, logs and samples. [10]

Week 5 checkpoint

Transit centers align across sectors, posterior summaries are stable across runs, and residual checks support the adopted model. Investigate any persistent timing pattern or correlated residuals before interpreting the intervals.

Measure and explain uncertainty

Report the posterior interval under the adopted fit and show how key estimates move under reasonable analysis choices.

1. Summarize the posterior

For each parameter report the median and the central 68% credible interval from the 16th and 84th percentiles. The lower error is median minus P16 and the upper error is P84 minus median. Derive quantities draw by draw to retain correlations. If reporting physical planet radius, multiply each radius-ratio draw by a compatible stellar-radius draw and state the stellar-radius source and any independence assumption.

2. Test for sensitivity to processing and noise

Change one choice at a time. Compare a transit-masked detrending window with a wider one, or compare against a local baseline fit on unflattened PDCSAP. Check residual scatter as a function of bin size through transit timescales. If it remains correlated, test a correlated-noise model. With enough data, refit without each sector in turn. Record median shifts and changes in interval width for each variant.

3. Check signal recovery

Inject several synthetic transits into suitable out-of-transit data at different phases, using the actual cadence and gaps. Reapply the same cleaning, transit masks, detrending and fit. Compare the injected radius ratio and timing with the recovered values and intervals. Use these trials to identify bias or underestimated errors. Flag any sensitivity comparable to or larger than the baseline interval.

4. Compare like with like

Match the parameter definition, units and passband where relevant. Compare epochs at the same transit number. Propagate each posterior draw with $T(N) = T0 + N \times P$ so timing and period covariance is retained. If the publication lacks covariance, state the assumption used in propagation. Flag shared TESS sectors or informative priors and report shifts descriptively when the estimates are dependent.

5. Publish a compact comparison table

Use one row per target and parameter: **parameter and unit | literature value and errors | reference | new median and 68% interval | difference | sensitivity and overlap notes**. Prioritize radius ratio, period and transit epoch. Include depth only if its definition and bandpass are comparable. For missing archive errors, report your fit interval and mark the published uncertainty as unavailable.

Week 8 checkpoint

Deliver the comparison table, phase-folded fit, per-sector residuals, corner plot and sensitivity results. Document which uncertainty sources were included in the model and which were explored by rerunning it.

Extend and package the project

The eight-week core delivers one complete target. Use the third month for a second well-observed planet or stronger checks on the first.

WEEK 9	**Choose the extension** Select a second target using the same screening criteria, or reserve additional sectors of the primary target for a holdout check.
WEEK 10	**Repeat the fit** Run the same workflow on the second target or test how well the original fit predicts the held-out sectors.
WEEK 11	**Check robustness** Expand the injection-recovery trials or test the effect of limb darkening and noise assumptions on the main results.
WEEK 12	**Archive the analysis** Reproduce the final figures and tables from saved inputs. Record software versions, seeds, exclusions and the exact reference rows.

Reproducibility checklist

Include the TAP query and archive snapshot, MAST product manifest, preprocessing notebook, versioned Python environment, per-run CSV configurations, posterior samples, diagnostics, comparison table and a README with rerun instructions.

Sources and setup references

[01] NASA Exoplanet Archive · Planetary Systems column definitions
[02] NASA Exoplanet Archive · TAP query guide
[03] Lightkurve · MAST light-curve search API
[04] NASA Exoplanet Archive · WASP-121 system overview
[05] NASA TESS Science Support Center · Working with TESS light curves
[06] NASA TESS Science Support Center · PDCSAP crowding corrections
[07] allesfitter · Photometry fit tutorial and input files
[08] allesfitter · Settings and exposure integration
[09] allesfitter · Fit parameters and priors
[10] allesfitter · Fitting and diagnostic API
[11] Günther and Daylan · allesfitter methods paper